EP8199 - GLOBAL OPTIMIZATION OF MIXED-INTEGER AND NON-CONVEX PROGRAMS

COLLOQUIUM 1

Avinash S. R. Subramanian, Truls Gundersen, Paul I. Barton

NTNU, Department of Energy and Process Engineering

MIT, Department of Chemical Engineering

Outline

- CONVEX VS. NONCONVEX PROGRAMS
- CLASSIFICATION OF OPTIMIZATION PROBLEMS
- TYPES OF OPTIMA
- PROCEDURE FOR GLOBAL OPTIMIZATION
- LOWER BOUNDING TECHNIQUES
- MISCELLANEOUS
- SUPPLEMENTARY SLIDES

Basic Concepts I

Convex vs. Non-convex sets

Definition (Convex set)

A set C is convex if the line segment joining any two points in the set is entirely contained in the set, i.e., for any two points $\mathbf{x}_1$, $\mathbf{x}_2 \in C$, all the points $\mathbf{z}$ are also in the set, where:

$$\mathbf{z} \equiv \lambda \mathbf{x}_1 + (1-\lambda)\mathbf{x}_2, \quad \forall \lambda \in [0,1] \tag{1}$$

A non-convex set has at least one line segment where this property doesn't hold

Basic Concepts II

Convex vs Non-convex Functions

Definition (Convex function)

A function $f : C \to \mathbb{R}$ is convex if its domain C is a convex set and the function value evaluated at any point on the line segment joining any two points $\mathbf{x}_1$, $\mathbf{x}_2 \in C$ lies below the line segment joining the function values evaluated at these two points (the secant line), i.e.:

$$f(\lambda \mathbf{x}_1 + (1-\lambda)\mathbf{x}_2) \leq \lambda f(\mathbf{x}_1) + (1-\lambda) f(\mathbf{x}_2), \quad \forall \lambda \in [0,1], \quad \forall \mathbf{x}_1, \mathbf{x}_2 \in C \tag{2}$$

Basic Concepts III

Convex vs. Non-convex Programs

Definition (Convex Program)

Let $f : D \to \mathbb{R}$. Then (OP) is is a convex optimization problem if **both** the **feasible set** D is a convex set and the **objective function** f is a convex function:

$$\min_{\mathbf{x} \in D} f(\mathbf{x}) \tag{OP}$$

If either the set D is non-convex or the function f is non-convex the program is non-convex.

Problem types I

$$\begin{aligned} \min_{\mathbf{x},\mathbf{y}} \quad & f(\mathbf{x},\mathbf{y}) \\ \text{s.t.} \quad & \mathbf{g}(\mathbf{x},\mathbf{y}) \leq \mathbf{0} \\ & \mathbf{x} \in X \subset \mathbb{R}^{n_x}, \mathbf{y} \in Y \equiv \{0,1\}^{n_y} \end{aligned} \tag{GP}$$

where:

- $\mathbf{x}$ and $\mathbf{y}$ denote the continuous and binary decision variables respectively
- the set $U \equiv [0,1]^{n_y}$ is the convex hull of the set Y (i.e., the smallest convex set that contains every point in Y
- $f : X \times U \to \mathbb{R}$ is the continuous objective function, and $\mathbf{g} : X \times U \to \mathbb{R}^m$ corresponds to continuous functions defining the constraints

Problem types II

- **Linear Programming (LP)** - if there are no binary decision variables $\mathbf{y}$ and the functions f and $\mathbf{g}$ are linear
- **Mixed-Integer Linear Programming (MILP)** - if the functions f and $\mathbf{g}$ are linear.
- **Convex Nonlinear Programming** - if there are no binary decision variables $\mathbf{y}$, f and $\mathbf{g}$ are convex functions.
- **Mixed-Integer Convex Programming (MICP)** - if the functions f and $\mathbf{g}$ are convex functions
- **Nonconvex Nonlinear Programming (NLP)** - if there are no binary decision variables $\mathbf{y}$ and atleast one of the participating functions is nonconvex.
- **Nonconvex Mixed-Integer Nonlinear Programming (MINLP)** - if atleast one of the participating functions is nonconvex.

Local vs. Global Minimum I

NTNU

Definition (Global Minimum)

A point $\mathbf{x}^* \in D$ is a (global) *minimum* of f on D if:

$$f(\mathbf{x}) \geq f(\mathbf{x}^*), \ \forall \mathbf{x} \in D,$$

i.e., a minimum is a feasible point whose objective function value is less than or equal to the objective function value of all other feasible points.

Local vs. Global Minimum II

NTNU

Definition (Local Minimum)

A point $\mathbf{x}^* \in D$ is a *local minimum* of f on D if there exists an $\varepsilon > 0$ such that

$$f(\mathbf{x}) \geq f(\mathbf{x}^*), \ \forall \mathbf{x} \in D \cap N_\varepsilon(\mathbf{x}^*)$$

i.e., a local minimum is a feasible point whose objective function value is less than or equal to the objective function value of all other feasible points that are also in an ε neighborhood of the local minimum.

Exercise : Based on the above definition, determine if MILPs have local minima.
Exercise : **Local** vs **Global information**. Give one well-known example of each.

Convex Problems

Theorem (Convex Problems: Local optimality implies global optimality)

Suppose $\mathbf{x}^*$ is a local minimum of a convex optimization problem. Then $\mathbf{x}^*$ is also a global minimum.

Lower and Upper Bounding

The general procedure is to determine a sequence of **Lower Bounds LBDs** and **Upper Bounds UBDs** that bracket the optimal value. Converge the best known UBD and LBD such that the gap **UBD - LBD** is within a tolerance. Solvers usually report this gap.

Upper Bounding Techniques:

1. Determine any **feasible** point. Value of objective provides an UBD.
2. **Restriction**: Shrink the feasible set of an optimization problem by constraining it further. The optimal value or value or objective function value at any feasible point yield an UBD for the original optimization problem.

Lower Bounding Techniques:

1. **Relaxation** of constraints
2. **Convexification**
3. Using **Duality theory**
4. **Interval Analysis**

1. Relaxation of Constraints

Enlarging the feasible set of an optimization problem by relaxing constraints yields a lower bound. This is termed the Lower Bounding Principle:

Theorem (Lower Bounding Principle)

Suppose we have sets D and $E \subset \mathbb{R}^n$ such that $D \subset E$. Then

$$\min\{f(\mathbf{x}) : \mathbf{x} \in E\} \leq \min\{f(\mathbf{x}) : \mathbf{x} \in D\}$$

2. Convexification

This is the procedure of formulating an convex optimization problem that provides a LBD for a nonconvex problem

Definition (Convex Underestimator and Concave Overestimator of a function)

Consider a function $f : C \to \mathbb{R}$ defined on a convex set C. A convex function $f^{cv} : C \to \mathbb{R}$ and a concave function $f^{cc} : C \to \mathbb{R}$ are a convex underestimator and concave overestimator respectively of f on C if:

$$f^{cv}(\mathbf{x}) \leq f(\mathbf{x}), \quad \forall \mathbf{x} \in C$$
$$f^{cc}(\mathbf{x}) \geq f(\mathbf{x}), \quad \forall \mathbf{x} \in C$$

Together, these two functions are also referred to as convex and concave relaxations of a function. When these functions are the tightest possible relaxations, they are termed convex and concave envelopes.

- Library of convex relaxations for common univariate and bivariate functions. Active research area: Show Kannan Pg. 133.
- Building blocks to construct convex relaxations of more complicated functions.

Convexification - Auxiliary Variables and Constraints Method (AVM) I

NTNU

Constructing a convex programming relaxation for the following optimization problem:

$$\begin{aligned} \min_{x_1,x_2,x_3} \quad & f(x_1,x_2,x_3) = x_1 \cdot x_2 \cdot (5 + x_3^2) \\ \text{s.t.} \quad & x_1,x_2,x_3 \in [0,1] \end{aligned} \tag{3}$$

1. Break up the participating functions into a sequence of univariate and bivariate functions (also termed the factorable representation) by defining auxiliary variables. Determine the upper and lower bounds for the auxiliary variables based on the corresponding bounds for the original variables.

$$\begin{aligned} v_1 &= x_3^2 && \in [0,1] \\ v_2 &= 5 + v_1 && \in [5,6] \\ v_3 &= x_1 \cdot x_2 && \in [0,1] \\ v_4 &= v_2 \cdot v_3 && \in [0,6] \end{aligned}$$

Convexification - Auxiliary Variables and Constraints Method (AVM) II

NTNU

2. Formulate an equivalent optimization problem in terms of these introduced variables and equality constraints:

$$
\begin{aligned}
\min_{x_1,x_2,x_3,v_1,v_2,v_3,v_4} \quad & v_4 \\
\text{s.t.} \quad & v_1 = x_3^2 \\
& v_2 = 5 + v_1 \\
& v_3 = x_1 \cdot x_2 \\
& v_4 = v_2 \cdot v_3 \\
& x_1, x_2, x_3 \in [0,1], v_1 \in [0,1], v_2 \in [5,6], v_3 \in [0,1], v_4 \in [0,6]
\end{aligned}
\tag{4}
$$

3. The above problem is still nonconvex as a result of the presence of nonlinear equality constraints. Thus, the last step is to replace each nonlinear equality constraint with a pair of inequality constraints in terms of their corresponding convex underestimators and concave overestimators:
   - For $v_1 = x_3^2$, it is known that $v_1^{cv} = x_3^2$ and $v_1^{cc} = x_3$ over $[0,1]$, Thus: $v_1 \geq x_3^2$ and $v_1 \leq x_3$
   - Function v_2 is affine thus convex and doesn't require reformulation

Convexification - Auxiliary Variables and Constraints Method (AVM) III

- For $v_3 = x_1 \cdot x_2$, the well-known McCormick envelopes can be used (where the superscripts L and U denote the lower and upper bound of the corresponding variable as presented in Problem 4):

  $v_3^{cv} = \max\{x_1^L \cdot x_2 + x_2^L \cdot x_1 - x_1^L \cdot x_2^L,\ \ x_1^U \cdot x_2 + x_2^U \cdot x_1 - x_1^U \cdot x_2^U\}$

  $v_3^{cc} = \min\{x_1^L \cdot x_2 + x_2^U \cdot x_1 - x_1^L \cdot x_2^U,\ \ x_1^U \cdot x_2 + x_2^L \cdot x_1 - x_1^U \cdot x_2^L\}$

  This gives the following constraints after substitution of bounds:

  $v_3 \geq 0$

  $v_3 \geq x_2 + x_1 - 1$

  $v_3 \leq x_1$

  $v_3 \leq x_2$

- For $v_4 = v_2 \cdot v_3$, a similar procedure using McCormick envelopes gives:

  $v_4 \geq 5 \cdot v_3$

  $v_4 \geq 6 \cdot v_3 + v_2 - 6$

  $v_4 \leq 5 \cdot v_3 + v_2 - 5$

  $v_4 \leq 6 \cdot v_3$

Convexification - Auxiliary Variables and Constraints Method (AVM) IV

4. Substituting all the above inequalities yields Problem 5 which is a convex program:

$$
\begin{aligned}
\min_{x_1,x_2,x_3,v_1,v_2,v_3,v_4} \quad & v_4 \\
\text{s.t.} \quad & v_1 \geq x_3^2 \\
& v_1 \leq x_3 \\
& v_2 = 5 + v_1 \\
& v_3 \geq 0 \\
& v_3 \geq x_2 + x_1 - 1 \\
& v_3 \leq x_1 \\
& v_3 \leq x_2 \\
& v_4 \geq 5 \cdot v_3 \\
& v_4 \geq 6 \cdot v_3 + v_2 - 6 \\
& v_4 \leq 5 \cdot v_3 + v_2 - 5 \\
& v_4 \leq 6 \cdot v_3 \\
& x_1, x_2, x_3 \in [0,1], v_1 \in [0,1], v_2 \in [5,6], v_3 \in [0,1], v_4 \in [0,6]
\end{aligned}
\tag{5}
$$

Convexification - Auxiliary Variables and Constraints Method (AVM) V

NOTE:

- Using the AVM approach, the convex programming relaxation is typically of a much higher dimension.
- The optimal objective function value of the convex programming relaxation (e.g., Problem 5) is a lower bound on the optimal objective function value of the nonconvex program (e.g., Problem 3).

3. Using Duality Theory I

Consider the NLP Problem NLP-PR which is also termed the primal.

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{g}(\mathbf{x}) \leq \mathbf{0} \\ & \mathbf{x} \in X \subset \mathbb{R}^{n_x} \end{aligned} \tag{NLP-PR}$$

Definition (Lagrange function for Problem NLP-PR)

The Lagrange function (L) adds the objective function with a weighted sum of the constraint functions i.e., $L : \mathbb{R}^{n_x} \times \mathbb{R}^m \to \mathbb{R}$ is given by:

$$L(\mathbf{x}, \boldsymbol{\mu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^{\mathrm{T}} \mathbf{g}(\mathbf{x}) \tag{6}$$

Each element of the vector $\boldsymbol{\lambda}$ (termed a Lagrange multiplier) corresponds to a constraint.

3. Using Duality Theory II

NTNU

Definition (The Dual function for Problem NLP-PR)

The dual function q is defined implicitly as the optimal objective function value of an optimization problem parameterized by a Lagrange multiplier vector, i.e., $q : \mathbb{R}^m \to \mathbb{R}$ is given by:

$$q(\boldsymbol{\lambda}) = \inf_{\mathbf{x} \in X} L(\mathbf{x}, \boldsymbol{\lambda}) \tag{7}$$

We note that q is only defined on $\boldsymbol{\lambda}$ for which the optimization problem above is bounded.

Definition (The Dual of Problem NLP-PR)

The Dual is an optimization problem defined as:

$$\begin{aligned} \sup_{\boldsymbol{\lambda} \in \mathbb{R}^m} \quad & q(\boldsymbol{\lambda}) \\ \text{s.t.} \quad & \boldsymbol{\lambda} \geq \mathbf{0}, \end{aligned} \tag{NLP-DP-DEF}$$

3. Using Duality Theory III

The above definitions are combined to write out an alternative concise Dual Problem (Problem NLP-DP)

Definition (Dual - concise definition)

$$\sup_{\boldsymbol{\lambda} \geq \mathbf{0}} \quad \inf_{\mathbf{x} \in X} f(\mathbf{x}) + \boldsymbol{\lambda}^{\mathrm{T}} \mathbf{g}(\mathbf{x}) \tag{NLP-DP}$$

Definition (A Duality Multiplier of Problem NLP-PR)

A vector $\hat{\boldsymbol{\lambda}}$ is said to be a Duality Multiplier of Problem NLP-PR if:

$$\hat{\boldsymbol{\lambda}} \geq \mathbf{0} \quad \text{and} \quad f(\hat{\mathbf{x}}) = \inf_{\mathbf{x} \in X} f(\mathbf{x}) + (\hat{\boldsymbol{\lambda}})^{\mathrm{T}} \mathbf{g}(\mathbf{x}) \tag{8}$$

where $\hat{\mathbf{x}}$ in this case denotes the globally optimal solution. We note that the duality multiplier is not the same as a Lagrange multiplier and may not exist for certain problem classes.

3. Using Duality Theory IV

Theorem (Weak duality)

For all problem classes, the supremum on the Dual Problem NLP-DP is a lower bound on the optimal objective function value of the primal NLP (Problem NLP-PR).

Theorem (Strong duality and existence of duality multiplier for convex programs)

Suppose Problem NLP-PR is a convex program and has a finite solution value. Also suppose a strictly feasible point exists (also called a Slater point - a feasible point where none of the constraints are active). Then strong duality holds and Problems NLP-DP and NLP-PR have the same optimum objective function value. In addition, atleast one duality multiplier exists.

Why notions of open, closed set, boundedness were introduced

1. Differentiability is usually defined on an open set
2. **Existence of minima: Weierstrass theorem**

Theorem (Weierstrass Theorem)

If $D \subset \mathbb{R}^n$ is a nonempty compact set and $f : D \to \mathbb{R}$ is continuous on D, then both a minimum and a maximum of f on D exist.

ADDITIONARY MATERIAL

Convex & Differentiable objective function

NTNU

Theorem

Let $X \subset \mathbb{R}^n$ be an open set, let $C \subset X$ be a nonempty convex set and let $f : X \to \mathbb{R}$ be differentiable on X. If f is convex on C and for $\mathbf{x}^* \in C$

$$\langle \boldsymbol{\nabla} f(\mathbf{x}^*), (\mathbf{x} - \mathbf{x}^*) \rangle \geq 0, \quad \forall \mathbf{x} \in C$$

then $\mathbf{x}^*$ is a global minimum.

- This is a necessary and sufficient condition for (global) optimality. However, it is not convenient as we would like to explicitly consider the equalities and inequalities defining the constraints, as we do in the next sections.

Unconstrained Optimization Problems I

NTNU

Definition (Unconstrained Optimization Problem)

Let $f : \mathbb{R}^n \to \mathbb{R}$. Then (UOP) is is an *unconstrained optimization problem*:

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) \tag{UOP}$$

We can also equivalently write the UOP in terms of optimization on an open set $X \subset \mathbb{R}^n$

Theorem (1: Unconstrained, Necessary Condition if f is differentiable)

Let $X \subset \mathbb{R}^n$ be a nonempty open set and suppose that $f : X \to \mathbb{R}$ is differentiable at $\mathbf{x}^* \in X$:

1. If $\mathbf{x}^*$ is a local minimum of f on X then $\boldsymbol{\nabla} f(\mathbf{x}^*) = \mathbf{0}$.
2. If $\mathbf{x}^*$ is a local maximum of f on X then $\boldsymbol{\nabla} f(\mathbf{x}^*) = \mathbf{0}$.

- This condition only uses local (i.e., derivative information)
- Points $\mathbf{x}^*$ such that $\boldsymbol{\nabla} f(\mathbf{x}^*) = \mathbf{0}$ are known as *stationary points*. All local minima (and maxima) are stationary points but not all stationary points are local minima.

Unconstrained Optimization Problems II

NTNU

Theorem (2: Unconstrained, Necessary Condition if f is twice differentiable)

Let $X \subset \mathbb{R}^n$ be a nonempty open set and suppose that $f : X \to \mathbb{R}$ is twice differentiable at $\mathbf{x}^* \in X$. If $\mathbf{x}^*$ is a local minimum of f on X then $\boldsymbol{\nabla} f(\mathbf{x}^*) = \mathbf{0}$ and $\boldsymbol{\nabla}^2 f(\mathbf{x}^*)$ is positive semidefinite.

Theorem (3: Unconstrained, Sufficient Condition if f is twice differentiable)

Let $X \subset \mathbb{R}^n$ be a nonempty open set and suppose that $f : X \to \mathbb{R}$ is twice differentiable at $\mathbf{x}^* \in X$. If $\boldsymbol{\nabla} f(\mathbf{x}^*) = \mathbf{0}$ and $\boldsymbol{\nabla}^2 f(\mathbf{x}^*)$ is positive definite then $\mathbf{x}^*$ is a strict local minimum of f on X.

Theorem (4: Unconstrained, Sufficient Condition if f is differentiable and convex)

Let $C \subset \mathbb{R}^n$ be a nonempty open convex set. Let $f : C \to \mathbb{R}$ be convex and differentiable on C. Then, $\mathbf{x}^* \in C$ is a (global) minimum of f on C iff $\boldsymbol{\nabla} f(\mathbf{x}^*) = \mathbf{0}$.

Unconstrained Optimization Problems III

NTNU

Theorem (5 (Optional - BSS 4.1.6.): Unconstrained, Sufficient Condition if f is univariate and infinitely differentiable)

Let the univariate function $f : \mathbb{R} \to \mathbb{R}$ be infinitely differentiable at $x^* \in \mathbb{R}$. Then x^* is a local minimum of f on $\mathbb{R}$ iff:

1. Either: $f^{(j)}(x^*) = 0$ for all $j = 1, 2, ...,$
2. Or: There exists an even $n \geq 2$ such that $f^{(n)}(x^*) > 0$ and $f^{(j)}(x^*) = 0$ for all $1 \leq j < n$, where $f^{(j)}$ denotes the jth-order derivative of f.

Unconstrained Optimization Problems IV

NTNU

Example 1: f is convex & quadratic

Consider:

$$\min_{\mathbf{x} \in X} (x_1 - 1)^2 + (x_2 - 2)^2$$
$$X = \{\mathbf{x} \in \mathbb{R}^2 : x_1 > 0, x_2 > 0\}.$$

X is an open set and the gradient vector of the objective function is given by:

$$\boldsymbol{\nabla} f(\mathbf{x}) = \begin{bmatrix} 2(x_1 - 1) \\ 2(x_2 - 2) \end{bmatrix}.$$

$\mathbf{x}^* = (1, 2)$ is a stationary point because:

$$\boldsymbol{\nabla} f(\mathbf{x}^*) = \begin{bmatrix} 2(1 - 1) \\ 2(2 - 2) \end{bmatrix} = \mathbf{0}.$$

Unconstrained Optimization Problems V

This is the only stationary point in X.
The Hessian matrix is:

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

which has eigenvalues $\lambda_1 = +2$ and $\lambda_2 = +2$. Hence, the Hessian matrix is positive definite on X and f is strictly convex on X. Thus, $\mathbf{x}^*$ is a global minimum.

Exercise: Which of the theorems presented so far are required to make this conclusion?

Unconstrained Optimization Problems VI

NTNU

Example 2: f is a bilinear function

Consider:

$$\min_{\mathbf{x}\in\mathbb{R}^2} \; x_1 x_2.$$

The gradient vector of the objective function is given by:

$$\boldsymbol{\nabla} f(\mathbf{x}) = \begin{bmatrix} x_2 \\ x_1 \end{bmatrix}$$

The Hessian matrix for is:

$$\boldsymbol{\nabla}^2 f(\mathbf{x}) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

which has eigenvalues $\lambda_1 = +1$ and $\lambda_2 = -1$. Hence, the Hessian matrix is indefinite on $\mathbb{R}^2$. The only stationary point in $\mathbb{R}^2$ is $\mathbf{x}^* = \mathbf{0}$. This stationary point is neither a local minimum nor a local maximum. Such stationary points are called *saddle points*.

Exercise: What does this say about the bilinear function, i.e., is it convex or nonconvex?

Inequality Constrained Problems I

Definition (Inequality Constrained Optimization Problem)

Let $X \subset \mathbb{R}^n$ be a nonempty open set, and $f : X \to \mathbb{R}$ and $\mathbf{g} : X \to \mathbb{R}^m$. Then:

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{g}(\mathbf{x}) \leq \mathbf{0} \\ & \mathbf{x} \in X. \end{aligned} \tag{ICOP}$$

is an *inequality constrained optimization problem* with m inequality constraints.

Inequality Constrained Problems II

NTNU

Definition (Active and Inactive Inequality Constraints, Active Set)

- An inequality constraint $g(\mathbf{x}) \leq 0$ is *active* (or *binding*) at a point $\hat{\mathbf{x}}$ if $g(\hat{\mathbf{x}}) = 0$.
- If $g(\hat{\mathbf{x}}) < 0$, then the constraint g is *inactive* (or *nonbinding*) at $\hat{\mathbf{x}}$.
- If $g(\hat{\mathbf{x}}) > 0$, then the constraint is *violated* at $\hat{\mathbf{x}}$.
- For any $\hat{\mathbf{x}} \in X$ let the active set $A(\hat{\mathbf{x}})$ be the index set of the active inequality constraints at $\hat{\mathbf{x}}$, i.e.:

$$A(\hat{\mathbf{x}}) = \{i : g_i(\hat{\mathbf{x}}) = 0\}.$$

Motivation: We make the distinction between active and inactive inequality constraints because inequality constraints that are inactive at a candidate local optimum will also not be active on some neighborhood of this point as a result of continuity of the participating f and g functions. Thus, conditions for local optimality will not involve inactive constraints.
- Slides Still In Progress for Inequality Constrained and General Problems. KKT conditions.